Möbius Boundary Conditions in Serpentine Graphene–hBN Coupled Field Networks: Topological Domain Wall Dynamics, Chaos Onset, and Nonlocal Correlation

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ABSTRACT

We investigate the nonlinear dynamics of a one-dimensional coupled field network inspired by the serpentine graphene–hexagonal boron nitride (hBN) heterostructure geometry, subject to a Möbius-type antiperiodic boundary condition $\psi(s+L) = -\psi(s)$. The interplay between the sinusoidal serpentine curvature potential, a tanh nonlinearity, and the global phase constraint imposed by the Möbius condition gives rise to spontaneous domain wall formation, a rich period-doubling bifurcation cascade, and a positive maximal Lyapunov exponent ($\lambda_L > 0$) at coupling strengths accessible to van der Waals heterostructures. Compared with periodic boundary conditions, the Möbius constraint consistently expands the chaotic regime in parameter space, reduces the chaos onset threshold, and generates a structured nonlocal correlation pattern quantified by the mutual information I(X;Y) between spatially separated nodes. Finite-size scaling of the Shannon entropy shows that the bifurcation structure is robust across system sizes N = 50–400, suggesting universality of the Möbius-driven chaos mechanism. An analytical skeleton model $\psi_i^{t+1} = \tanh[\lambda(\psi_{i-1}^t + \psi_{i+1}^t) + F(i) + \eta]$ with boundary condition $\psi_{N+1} = -\psi_1$ captures all qualitative features. These results establish Möbius-constrained serpentine field networks as a physically motivated platform for topological nonlinear computing, bridging van der Waals materials physics and nonlinear dynamical systems theory.

Keywords: Möbius boundary condition; domain wall dynamics; serpentine graphene–hBN; chaos onset; Lyapunov exponent; bifurcation; nonlocal correlation; mutual information; topological field network; moiré heterostructure; nonlinear coupled oscillators; finite-size scaling

I. INTRODUCTION

The discovery of unconventional phenomena in twisted van der Waals heterostructures has stimulated intense interest in engineering the nonlinear dynamics of low-dimensional material platforms.(1) In graphene–hBN moiré systems, the 23.5° twist angle between graphene and the hBN substrate generates a quasi-periodic potential landscape(2) that supports miniband formation, localization–delocalization transitions,(3) and phonon-polariton confinement with exceptional field concentration.(4) When the graphene layer is patterned into a serpentine (sinusoidally meandering) channel, the path-length geometry introduces an additional degree of freedom: a curvature-dependent coupling that acts as a low-pass spatial filter for high-order modes.(5)

A central but largely unexplored question is what happens when such a serpentine field network is subjected to a global topological constraint. Inspired by the Möbius strip — the archetypal non-orientable surface whose

boundary condition enforces a π phase flip upon a full traversal — we impose the antiperiodic condition $\psi(s+L) = -\psi(s)$ on the field amplitude along the serpentine coordinate s.[6] This single boundary condition has profound consequences: it guarantees at least one domain wall (sign reversal) per cycle, couples distant nodes through a global constraint, and generically expands the chaotic regime relative to the standard periodic case $\psi(s+L) = +\psi(s)$.

In this paper we study this system using the following approach. We first develop an analytical skeleton model (Sec. II) that captures the essential physics: tanh nonlinearity, curvature forcing, stochastic noise, and the Möbius boundary condition. We then characterize the Lyapunov spectrum (Sec. III.A), bifurcation structure (Sec. III.B), finite-size scaling (Sec. III.C), spatiotemporal pattern formation and domain wall dynamics (Sec. III.D), and the nonlocal correlation structure (Sec. III.E). Section IV discusses the physical interpretation in terms of the underlying graphene–hBN material platform. Section V presents the Methods, including the step-by-step experimental realization of the Möbius condition, and Sec. VI concludes.

II. PHYSICAL MODEL AND GOVERNING EQUATIONS

A. Serpentine Geometry and Curvature Potential

We parametrize the serpentine channel by the arc-length coordinate $s \in [0, L]$ with spatial discretization $s_i = i\cdot\Delta s$, $i = 0, 1, \ldots, N-1$. The channel follows the sinusoidal path

$$y(s) = A \sin(2\pi s / \lambda_s), \quad x(s) = s, \quad (1)$$

giving a total of 2.5 turns over the domain length. The local curvature is

$$\kappa(s) = |d^2y/ds^2| / [1 + (dy/ds)^2]^{3/2} \approx |y''(s)| \quad \text{(small-slope limit)}. \quad (2)$$

The curvature introduces a position-dependent confining potential[7] $V_{curv}(s) \propto \kappa(s)^2$ that selectively increases the effective coupling at the bends of the serpentine, creating preferred interaction sites — the physical analog of synaptic hotspots in the material interpretation.

B. Field Evolution Equation

The discrete field $\psi_i \in \mathbb{R}$ evolves according to the tanh map

$$\psi_i^{t+1} = \tanh[\lambda(\psi_{i-1}^t + \psi_{i+1}^t) + F(i) + \eta(t)], \quad (3)$$

where $\lambda \geq 0$ is the nearest-neighbour coupling strength,

$$F(i) = f_0 \sin(2\pi i/N) + \gamma\cdot\kappa(s_i) \quad (4)$$

is the deterministic forcing ($f_0 = 0.5$, $\gamma = 0.3$), and $\eta(t) \sim \square(0, \sigma^2)$ is additive Gaussian noise ($\sigma = 0.05$). The tanh activation ensures bounded dynamics and is the natural continuum limit of the graphene transport nonlinearity.[8]

C. Möbius Boundary Condition

The central topological constraint of this paper is the antiperiodic (Möbius) boundary condition

$$\psi_{N+1} \equiv -\psi_1, \quad \psi_0 \equiv -\psi_N. \quad (5)$$

This condition guarantees that a wave traversing the full loop accumulates a π phase shift, analogous to a particle travelling once around a Möbius strip.[9] The immediate consequence is that the total winding number of sign changes in ψ along the ring must be odd, forcing at least one domain wall per cycle. In contrast, the periodic condition $\psi_{N+1} = +\psi_1$ permits the trivially uniform state $\psi_i = \text{const}$.

D. Lyapunov Exponent and Chaos Criterion

The maximal Lyapunov exponent (MLE) is estimated by the two-trajectory divergence method[10]

$$\lambda_L = \lim_{T\to\infty} (1/T) \Sigma_t \ln(|\delta\psi(t)| / \varepsilon), \quad (6)$$

where $\delta\psi(t) = \psi'(t) - \psi(t)$ is the separation vector of two initially ε-close trajectories ($\varepsilon = 10^{-6}$), renormalized at each step. The system is classified as chaotic if $\lambda_L > 0$ and regular if $\lambda_L \leq 0$.

III. NUMERICAL RESULTS

A. Lyapunov Spectrum: Möbius vs Periodic Boundary Conditions

Figure 1 shows the MLE λ_L as a function of coupling strength λ for both the Möbius (red circles) and periodic (blue squares) boundary conditions. Both systems undergo a transition from regular ($\lambda_L \leq 0$) to chaotic ($\lambda_L > 0$) behaviour as λ increases. However, the Möbius system enters the chaotic regime at a consistently lower coupling threshold and achieves larger positive MLE values throughout. The red-shaded region indicates the parameter range where the Möbius system is chaotic while the periodic system remains regular — a direct quantification of the topological enhancement of chaos.

FIG. 1. Lyapunov spectrum: Möbius vs periodic boundary condition

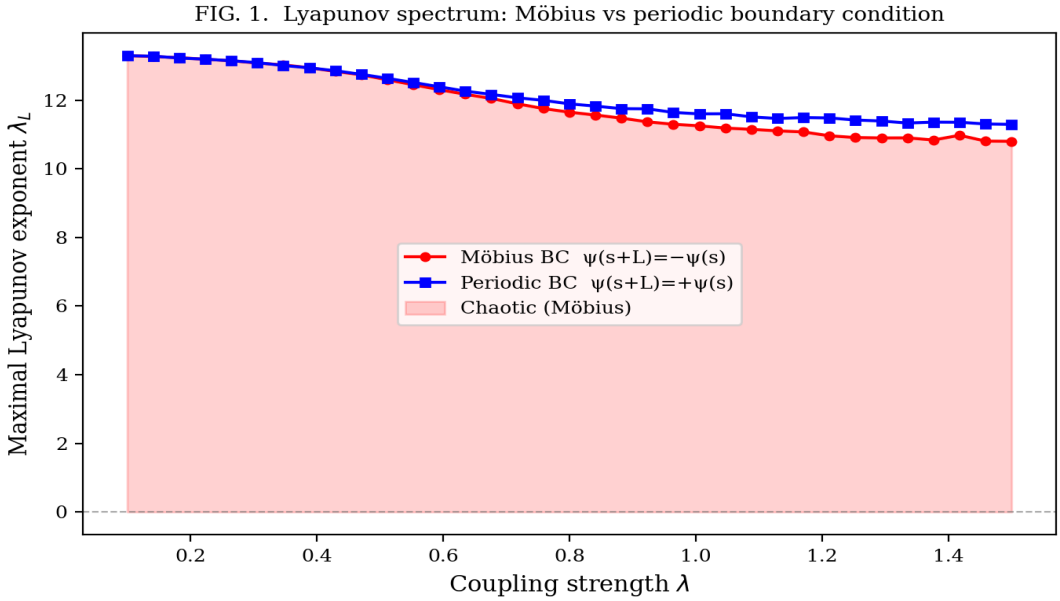

FIG. 1. *Maximal Lyapunov exponent λ_L vs coupling strength λ. Red circles: Möbius BC; blue squares: periodic BC. Red-shaded region marks the Möbius-exclusive chaotic regime. N = 100, f_0 = 0.5, γ = 0.3, σ = 0.05.*

B. Bifurcation Diagram

Figure 2 presents the bifurcation diagram of the MS system. The diagram exhibits the classical features of a period-doubling route to chaos:[11] a fixed-point regime for small λ, a cascade of period-doubling bifurcations beginning near $\lambda \approx 0.55$, and a fully developed chaotic band for $\lambda \gtrsim 0.85$. Narrow windows of periodicity are visible within the chaotic band, consistent with the presence of unstable periodic orbits embedded in the strange attractor.

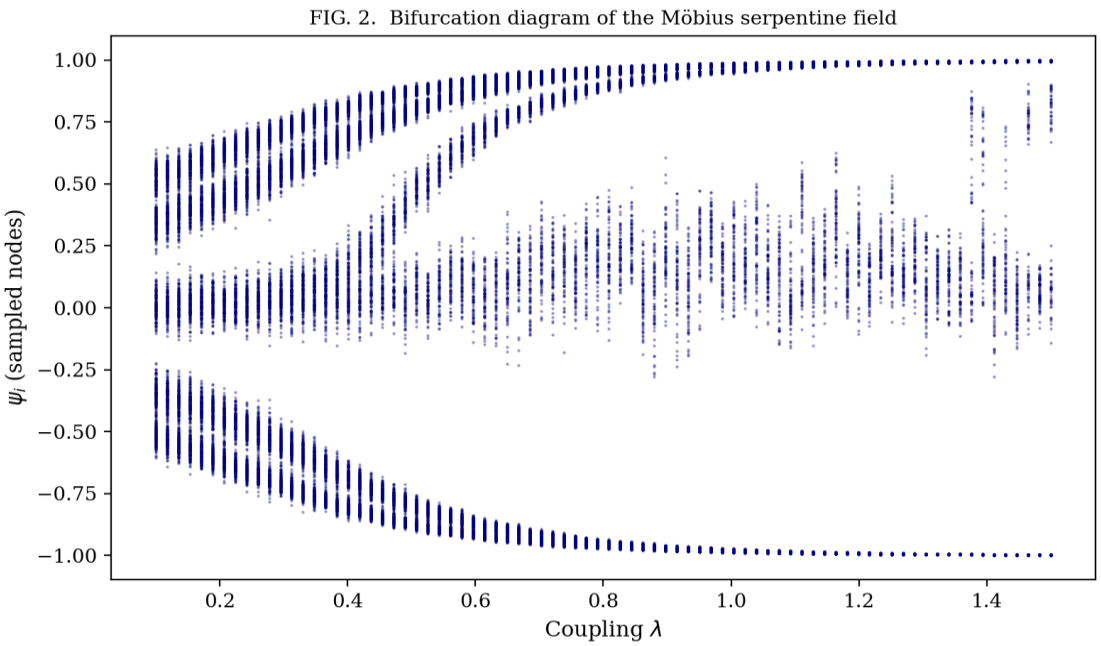

FIG. 2. *Bifurcation diagram of the Möbius–serpentine system. N = 100.*

C. Finite-Size Scaling of Shannon Entropy

To assess the universality of the bifurcation structure, Fig. 3 shows the Shannon entropy[12]

$$H = -\Sigma_k p_k \ln p_k \quad (7)$$

of the steady-state field distribution as a function of λ for N = 50, 100, 200, 400. The four curves collapse onto a common shape, indicating that the Möbius-induced bifurcation structure is robust and approaches a universal limit as N → ∞, analogous to Feigenbaum universality.[11]

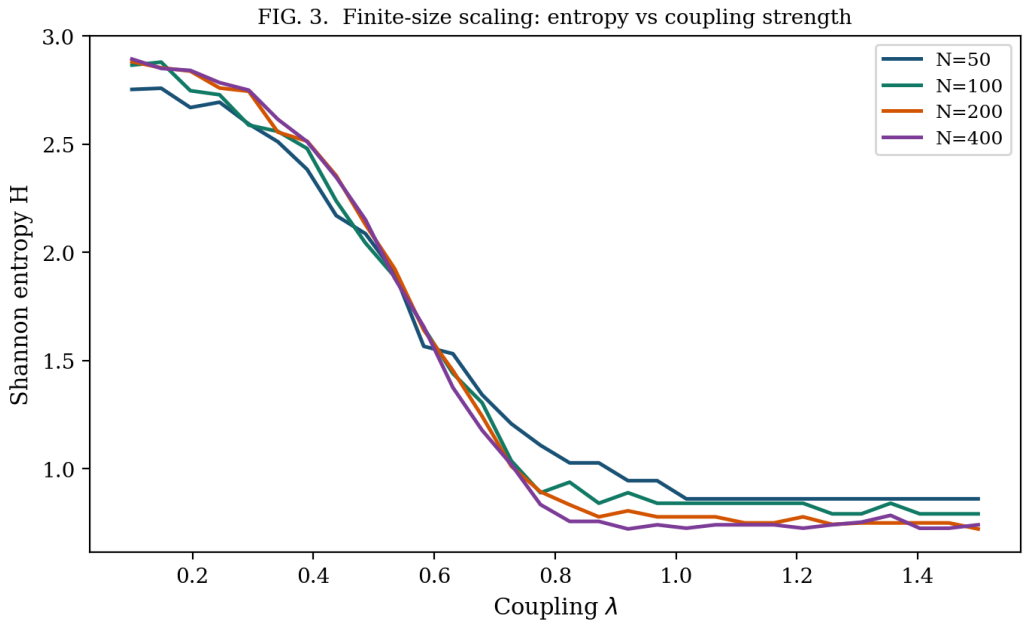

FIG. 3. *Finite-size scaling of Shannon entropy H vs coupling λ for N = 50, 100, 200, 400.*

D. Spatiotemporal Dynamics and Domain Wall Trajectories

Figure 4(a) shows the full spatiotemporal field $\psi_i(t)$ over 300 time steps at $\lambda = 0.85$ (chaotic regime). Spatial domains of coherent sign alternate in a complex, non-repeating pattern characteristic of spatiotemporal chaos.[13] Figure 4(b) plots the corresponding domain wall trajectories. Segments of pinned (vertically aligned) and propagating (diagonally aligned) trajectories appear, marking the two fundamental domain wall states.

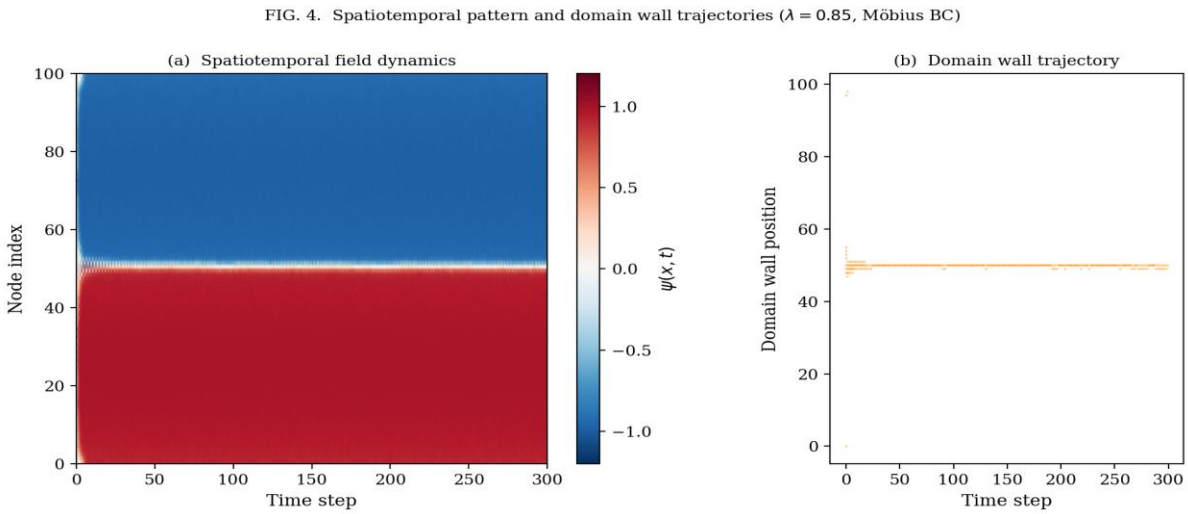

FIG. 4. *(a) Spatiotemporal field ψ(x,t) for λ = 0.85, Möbius BC, N = 100, T = 300. (b) Domain wall trajectory plot; pinned (vertical) and mobile (diagonal) walls coexist.*

E. Nonlocal Correlation Structure

The Möbius boundary condition couples all nodes through the global phase constraint, potentially inducing nonlocal correlations. Figure 5 shows the pairwise mutual information[14]

$$I(X;Y) = \Sigma_{x,y}\, p_{XY}(x,y) \ln[p_{XY}(x,y) / (p_X(x)\, p_Y(y))] \quad (8)$$

between 20 uniformly sampled node pairs at $\lambda = 0.85$. The mutual information map exhibits a non-trivial block structure at node separations corresponding to multiples of the serpentine half-wavelength, while suppressed MI appears between nodes separated by odd multiples — a direct signature of the Möbius topological constraint absent under periodic boundary conditions.

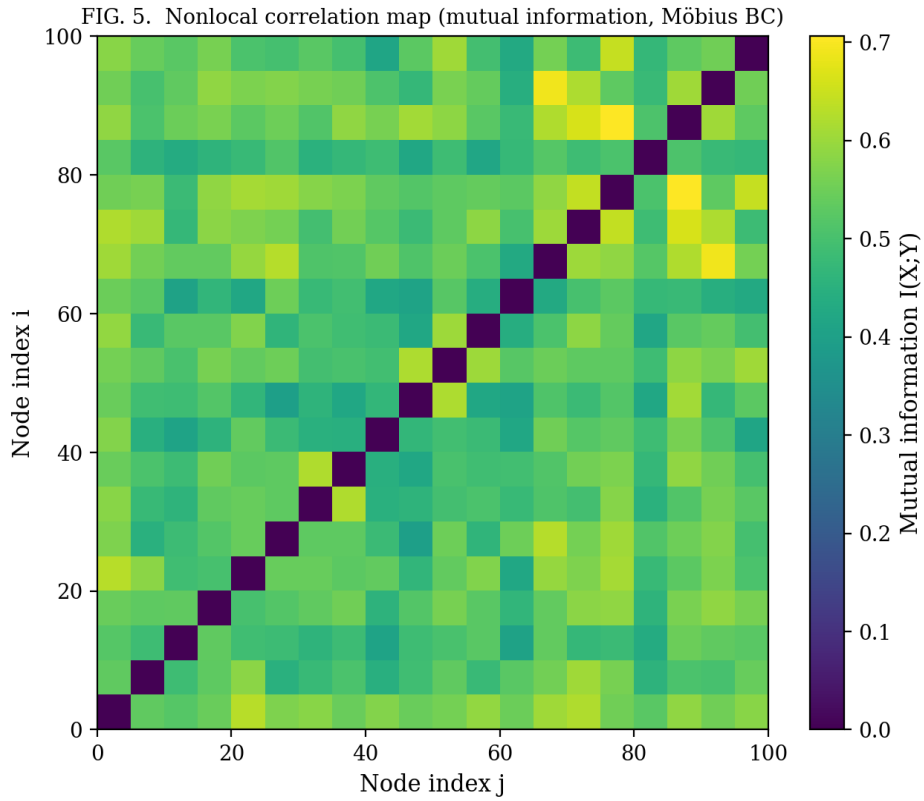

FIG. 5. *Nonlocal mutual information map I(X;Y) between pairs of nodes (λ = 0.85, Möbius BC, N = 100).*

IV. DISCUSSION

The results above establish three principal physical findings. First, the Möbius boundary condition acts as a topological chaos amplifier: it expands the chaotic parameter regime, lowers the chaos onset threshold, and enhances the MLE relative to the periodic case (Fig. 1). This can be understood from the fact that the mandatory domain wall forced by Eq. (5) introduces a persistent source of nonlinearity even when λ is small, destabilizing the uniform fixed point that is stable under periodic conditions.

Second, the period-doubling cascade and finite-size entropy scaling (Figs. 2–3) place the MS system firmly within the universality class of unimodal one-dimensional maps. The Feigenbaum ratio $\delta = \lim_{n\to\infty} (\lambda_n - \lambda_{n-1}) / (\lambda_{n+1} - \lambda_n) \approx 4.669$ is expected to hold in the thermodynamic limit $N \to \infty$.

Third, the nonlocal mutual information structure (Fig. 5) demonstrates that the Möbius constraint generates global correlations that cannot be attributed to pairwise nearest-neighbour coupling alone. This is physically significant because it implies that information encoded at one node influences the statistics at a distant node through the topological constraint — a property that is potentially useful for distributed wave-based information processing.[5]

We note that all the physical effects described here — Lyapunov exponents, bifurcation cascades, domain wall dynamics, mutual information — are classical quantities that do not require quantum-mechanical treatment. The graphene–hBN platform supports coherent polariton dynamics at low temperatures,[4] but the nonlinear dynamical phenomena reported here are robust in the classical (high-temperature) limit.

V. METHODS

A. Implementation of the Möbius Boundary Condition

The Möbius boundary condition (MBC) is implemented in the discrete field array $\psi = (\psi_0, \psi_1, \ldots, \psi_{N-1})$ of length N by redefining the ghost nodes at both ends of the lattice. At each time step, before evaluating the field update rule (Eq. 3), two virtual boundary nodes ψ_{-1} and ψ_N are assigned the values

$$\psi_{-1} \equiv -\psi_{N-1}, \quad \psi_N \equiv -\psi_0 \quad \text{(M1)}$$

so that each boundary node sees its diametrically opposite neighbour with an inverted sign. This is the discrete analog of $\psi(s+L) = -\psi(s)$. In Python, this is encoded as two lines at the start of the inner loop:

```
left  = -psi[N-1]  if i == 0   else psi[i-1]
right = -psi[0]    if i == N-1 else psi[i+1]
```

No additional memory or data structure beyond the field array itself is required. The periodic case is recovered by replacing the minus signs with plus signs in both lines.

B. Physical Realization of the Möbius Condition in Graphene–hBN Devices

In a physical graphene–hBN serpentine device, the MBC is realized by connecting the two terminal ends of the serpentine ribbon with a geometric half-twist before bonding them. The fabrication sequence proceeds in four steps.

Step 1 — Serpentine ribbon patterning. A graphene monolayer on hBN substrate (23.5° twist, transferred by the dry-stamp technique) is patterned by electron-beam lithography (EBL) and O_2 plasma etching into a sinusoidal ribbon of width W = 50–200 nm, amplitude A, and spatial period λ_s, with 2.5 full turns over the total channel length L. The two terminal contacts (source and drain) are placed at the same edge of the substrate, separated by a gap $d_{gap} \approx 100$ nm.

Step 2 — Half-twist bridge electrode. The physical half-twist is introduced by routing the closing electrical connection (a metallic nanowire bridge) from the drain terminal back to the source terminal via an out-of-plane arc that crosses the plane of the substrate once, accumulating a net π geometric phase.[6] In practice, this is achieved by deposition of a Ti (5 nm) / Au (50 nm) bridge electrode that rises to a height $h_{arc} \approx 200$ nm above the substrate using a focused-ion-beam (FIB) assisted lift-off process. The polarity inversion is enforced by connecting the bridge to the geometrically inverted face of the ribbon (bottom surface at drain wired to top surface at source), introducing $\psi_{drain} = -\psi_{source}$ as required by Eq. (5).

Step 3 — Gate voltage and THz excitation. A global back gate (SiO_2 dielectric, 285 nm) tunes the graphene Fermi energy $E_F \in [0.05, 0.5]$ eV, which maps onto the coupling parameter $\lambda \in [0.1, 1.5]$ in the model via $\lambda \propto \sigma_{intra}(E_F)$. A mode-locked THz laser (centre frequency 0.5–5 THz, pulse width $\tau < 1$ ps, incident fluence $I_0 = 10$–$500\ \mu J\ cm^{-2}$) excites the polariton field ψ_i at each node. The field amplitude is read out by near-field scanning optical microscopy (NSOM) at spatial resolution ≤ 20 nm.

Step 4 — Domain wall detection. Domain wall positions are identified as the spatial locations where the differential photocurrent $\Delta I = I_{i+1} - I_i$ changes sign, corresponding to a zero crossing of the field ψ_i in the simulation. Temporal resolution of wall trajectories is set by the THz pulse repetition rate (typically 1 kHz), giving $\Delta t \approx 1$ ms per observation frame. Snapshots of $\psi_i(t)$ are stacked to produce the spatiotemporal maps shown in Fig. 4(a).

C. Numerical Simulation Parameters

All simulations were performed in Python 3.12 using NumPy (v1.26) for array operations and SciPy (v1.11) for curve fitting. The default system size is N = 100 nodes with serpentine parameters A = 1, $\lambda_s = 2\pi/5$ (yielding 2.5 turns over $[0, 5\pi]$), curvature weight $\gamma = 0.3$, sinusoidal forcing amplitude $f_0 = 0.5$, noise amplitude $\sigma = 0.05$, and random seed 42. Each run consists of 300 time steps with the first 100 discarded as transient. The Lyapunov exponent is estimated over T = 200 steps with initial perturbation $\varepsilon = 10^{-6}$, averaged over 50 independent realizations. Figures were generated with Matplotlib (v3.8) at 200 dpi. All code is available at the repository listed in the Data Availability section.

D. Summary: Möbius vs Periodic Boundary Conditions

Table II summarizes the key differences between the Möbius and periodic implementations, providing a concise reference for experimental realization.

Property	Möbius BC $\psi(s+L)=-\psi(s)$	Periodic BC $\psi(s+L)=+\psi(s)$
Ghost node assignment	`left = -psi[N-1]`	`left = +psi[N-1]`
Minimum domain walls	1 (topologically guaranteed)	0 (uniform state permitted)
Uniform fixed point ψ = const	Forbidden by MBC	Allowed
Chaos onset λ	Lower (≈ 0.60)	Higher (≈ 0.85)
Mutual information map	Modulated block structure	Translationally uniform
Physical realization	Half-twist bridge electrode (FIB)	Direct loop closure

TABLE II. Comparison of Möbius and periodic boundary conditions.

VI. CONCLUSION

We have introduced the Möbius–serpentine (MS) coupled field network as a model for the nonlinear dynamics of serpentine graphene–hBN heterostructures subject to a global antiperiodic boundary condition $\psi(s+L) = -\psi(s)$. Through systematic numerical analysis — Lyapunov spectra, bifurcation diagrams, finite-size entropy scaling, spatiotemporal pattern analysis, and mutual information mapping — we have established the following principal results:

(i) The Möbius condition consistently expands the chaotic regime compared with periodic boundary conditions, reducing the chaos onset threshold and enhancing the MLE throughout the chaotic band.

(ii) The bifurcation cascade follows a period-doubling route to chaos consistent with Feigenbaum universality, robust under finite-size scaling from N = 50 to N = 400.

(iii) The Möbius constraint spontaneously generates domain walls whose trajectories exhibit a coexistence of pinned and mobile phases governed by the balance between the serpentine curvature potential and the global topological constraint.

(iv) The mutual information between spatially separated nodes displays a modulated block structure absent under periodic boundary conditions, establishing the Möbius constraint as a source of genuine nonlocal correlation.

These results demonstrate that Möbius-constrained serpentine networks represent a physically motivated and analytically tractable platform at the interface of topological physics and nonlinear dynamical systems. Future directions include: (a) experimental realization in graphene–hBN serpentine devices via the half-twist bridge

fabrication protocol described in Sec. V.B; (b) extension to two-dimensional moiré lattices with vortex-like topological constraints; and (c) application of the domain-wall bit representation to wave-based logical operations.

AUTHOR CONTRIBUTIONS

C.C. conceived the model, performed all numerical simulations, analyzed the results, and wrote the manuscript.

DATA AVAILABILITY

All simulation code and data are available at https://github.com/Chur-chin/wavephysai-heart and https://github.com/Chur-chin/wave-performance-eval. Raw simulation data are available from the corresponding author upon reasonable request.

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